Seizures

Is the Patient Currently Seizing?

NO

Maintain Airway / Breathing / Circulation

Place patient in recovery position if unresponsive

Consider OPA/NPA
if patient deeply unconscious

Check Blood Sugar
Correct hypoglycemia

Obtain IV Access

Consider Other Causes of Seizures:

- Meningitis
- Head injury
- Hypoglycemia
- Eclampsia
- Pyrexia
- Poisons
- Alcohol
- Drug withdrawal

Go to Appropriate CPG

Go to Hypoglycemia CPG

YES

Consider ALS Back-Up

Airway:
Consider OPA/NPA — EMR

Titrate Oxygen
To maintain Sp02 >94% — EMR

Check Blood Sugar
Correct hypoglycemia

Obtain IV Access — EMT

Administer Benzodiazepines — EP

	ADULT	PEDIATRIC
Midazolam IM	10mg (single dose)	0.2mg/kg (max 5mg)
Midazolam IV	5mg	0.1mg/kg (maximum 5mg)
OR		
Diazepam IV	5mg	0.1mg/kg (maximum 5mg)
Diazepam PR	-	0.5mg/kg (maximum 5mg)

If Seizure does not Stop
Repeat Benzodiazepine dose x1

Consider OPA/NPA
or advanced airway if continued seizure, or patient deeply unconscious

Patient Care Goals

1. Terminate active seizures or prevent recurrence in the prehospital setting to protect the patient from injury and hypoxia.
2. Ensure safe transport to an appropriate medical facility for further evaluation and management of underlying causes.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with seizures or post-ictal states, including:
  - Active generalized tonic-clonic seizures (e.g., convulsions, rhythmic jerking).
  - Focal seizures (e.g., localized twitching, sensory disturbances).
  - Status epilepticus (seizures lasting >5 minutes or recurrent seizures without recovery).
  - Post-ictal state (e.g., confusion, lethargy, or weakness following a seizure).
  - History of epilepsy, recent seizure activity, or precipitating factors (e.g., medication non-compliance, fever, or trauma).

Exclusion Criteria

- Patients without history or description of seizure activity.

Key Patient Assessment Points

1. **Seizure Characteristics**
   - Confirm active seizure activity (e.g., tonic-clonic movements, focal twitching, or staring spells).
   - Determine duration of seizure and whether it is ongoing or has stopped.
   - Note frequency and pattern (e.g., single seizure, recurrent, or status epilepticus).
2. **Seizure History**
   - Inquire about history of epilepsy, prior seizures, or anticonvulsant medication use.
   - Identify precipitating factors (e.g., missed medications, sleep deprivation, fever, alcohol withdrawal, or head trauma).
   - Check for medical alert bracelets or documentation indicating epilepsy.
3. **Post-Ictal Assessment**
   - Evaluate for post-ictal state (e.g., confusion, lethargy, or focal neurological deficits).
   - Assess for injuries sustained during seizure (e.g., tongue biting, head trauma, or fractures).
4. **Glucose Assessment**
   - Check random blood sugar (RBS) to rule out hypoglycemia (<70 mg/dL), which can trigger seizures or mimic seizure activity.
5. **Assessment for Non-Epileptic Causes**
   - Assess for non-epileptic causes of seizures (e.g. Meningitis, Head Injury, Hypoglycemia, Eclampsia, Pyrexia, Poisons, Alcohol, Drug withdrawal).
6. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority:
       - Ongoing active Seizures
       - Status Epilepticus

Adult Patient Management & Interventions

1. **Primary Survey**
   - Perform Primary and Secondary Survey as per Universal Care CPG.
   - **Administer Oxygen**
     - Administer to maintain SpO2 ≥94% during or after seizures to prevent hypoxia.
2. **Consider Red Flags**
   - Status epilepticus (seizures >5 minutes or recurrent without recovery).
   - Respiratory depression or hypoxia (SpO2 <90%) during or after seizure.

- Suspected head trauma or injury sustained during seizure.
- First-time seizure or no known epilepsy history.
- Pregnancy with seizures

3. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be routinely requested for active seizures.
     - Early administration of benzodiazepines for active seizures improves patient outcomes, it is appropriate to meet the ALS unit en route to hospital.
4. **Active Seizure Management**
   - **Protect the Patient**
     - Remove hazards, pad surroundings, and avoid restraining movements to prevent injury.
     - Position the patient in the recovery position post-seizure to protect airway
   - **Suction and Airway Support**
     - Suction oral secretions or vomit to maintain airway patency.
     - Use nasopharyngeal or oropharyngeal airway if needed in unresponsive patients.
   - **Establish IV Access**
     - Indicated for active seizures, status epilepticus, or anticipated medication administration.
     - Consider intraosseous (IO) access if IV access cannot be obtained
   - **Medication Administration**
     - **Midazolam**
       - Adults:
         - IM: 10mg, single dose, do not repeat.
         - IV: 5mg, repeat once after 5-10 minutes if needed. (maximum 10mg)
       - Pediatrics:
         - IM: 0.2mg/kg, (maximum 10mg) repeat once after 10-20 minutes if needed.
         - IV: 0.1mg/kg, (maximum 5mg) repeat once after 10-20 minutes if needed.
     - **Diazepam**
       - Adults:
         - IV: 5mg, repeat once after 5 minutes if needed (maximum 10mg.)
       - Pediatrics:
         - IV: 0.1mg/kg (maximum 5mg), repeat once after 5 minutes if needed.
         - PR: 0.5mg/kg (maximum 5mg), repeat once after 5 minutes if needed.
5. **Post Seizure/ Post Ictal Management**
   - Once seizures stop, airway management is extremely important.
     - Position the patient in the recovery position and provide oxygen to maintain Sp02 >94%.
     - Basic airway adjuncts (OPA/NPA) may be used if required.
     - Advanced airways are not commonly required unless airway compromise persists.
6. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - Ongoing Monitoring:
       - Continuous monitoring of vital signs post-seizure is mandatory, particularly for patients who have a reduced level of consciousness or have received benzodiazepines.

Patient Safety Considerations

- **Calling for Back-up**
  - ALS back-up should be routinely requested for any patient who is actively seizing.
  - If ALS back-up is not immediately available, ALS back-up should be met en route to hospital, to ensure the rapid administration of benzodiazepines and termination of seizure activity.
- **Pediatric Patients**
  - Consider febrile seizures in children <5 years; assess for fever and underlying infection.

Clinical Insights & Additional Guidance

- **Glucose Testing**
  - Glucose should be tested prior to the administration of benzodiazepines.
- **IM Diazepam**
  - Diazepam is not recommended to be given IM due to its slow onset of action. Use IM midazolam.
    - If diazepam is required to be administered IM:
      - Adults: 10mg IM
      - Pediatric: 0.2mg/kg IM (maximum 10mg)

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Seizure activity characteristics (e.g., tonic-clonic, focal, status epilepticus), duration, and frequency.
- Injuries / Information: Medical alert tag, trauma or injury sustained, medication history, and known seizure disorder.
- Signs: GCS or AVPU, post-ictal state findings, vital signs, pupil response, presence of hypoxia or cyanosis, glucose level.
- Treatment:
  - Time seizure started and stopped.
  - Benzodiazepines given: type, route, dose, time.
  - Airway interventions: suctioning, airway adjuncts used.
  - Oxygen therapy provided and SpO2 reading.
- Allergies: Any known drug allergies relevant to anticonvulsant therapy.
- Medications: List of home AEDs (anti-epileptic drugs), last dose taken, if known.
- Background: History of epilepsy, known seizure triggers, comorbidities, or new-onset seizures.
- Other:
  - Scene safety, presence of bystanders, witness account of seizure.
  - Coordination with ALS for medication support.

Key Performance Indicators

1. Administer benzodiazepines are administered for all patients in status epilepticus.
2. Request ALS back-up for all patients with active seizures.
3. Perform an RBS for all seizure patients.

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